Settimana: 18

Materia: Fisica
Classe: 5F
Data: 2/03/2026

Argomenti:

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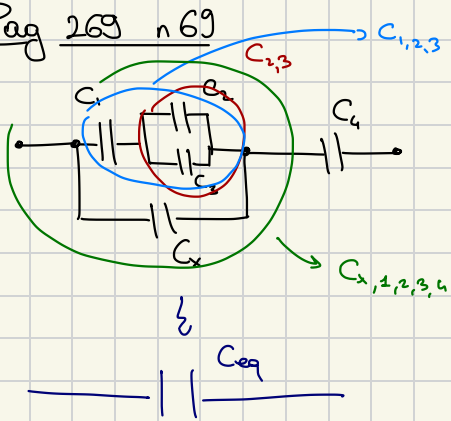


$C_{eq} = 2,4\,\mu F = 2,4 \cdot 10^{-6} F$

$C_1 = 2\,\mu F$
$C_2 = 1,25\,\mu F$
$C_3 = 1,75\,\mu F$
$C_4 = 6\,\mu F$

$C_x = ?$

$$C_{2,3} = C_2 + C_3 = 3\,\mu F$$

$$\frac{1}{C_{1,2,3}} = \frac{1}{C_1} + \frac{1}{C_{2,3}} \Rightarrow C_{1,2,3} = \frac{C_1 \cdot C_{2,3}}{C_1 + C_{2,3}} = \frac{6}{2+5}\,\mu F = \frac{6}{7}\,\mu F$$

$$C_{x,1,2,3} = C_x + C_{1,2,3} = C_x + \frac{6}{7}\,\mu F$$

$$C_{eq} = C_{x,1,2,3,4} = \frac{C_4\, C_{x,1,2,3}}{C_4 + C_{x,1,2,3}}$$

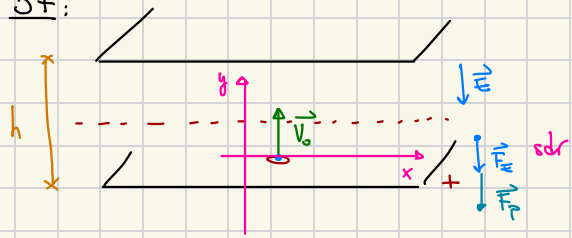
$$\frac{24}{10}\,\mu F = \frac{6\,\mu F \cdot \left(\frac{6}{7}\,\mu F + C_x\right)}{6\,\mu F + \frac{6}{7}\,\mu F + C_x} \Rightarrow \frac{\cancel{24}^{\,4}}{10}\left(6 + \frac{6}{7} + C_x\right) = \cancel{6}\left(\frac{6}{7} + C_x\right)$$

$$\rightsquigarrow \frac{2}{5}\,\frac{42+6+7C_x}{\cancel{7}} = \frac{6+7C_x}{\cancel{7}} \rightsquigarrow 36 + 14C_x = 30 + 35C_x$$

$$66 = 21\,C_x$$

$$\rightsquigarrow C_x = \frac{66}{21}\,\mu F$$

57:



$h = 4\ cm$
$A = 60\ cm^2$
$Q = 5{,}6\ nC$
Quanto vale V_0 affinché l'elettrone si fermi a metà dell'armature

È un moto accelerato uniforme

$$y(t) = y_0 + V_0 t + \frac{1}{2} a t^2$$

velocità al tempo 0

y_0 → spazio su y in 0

$$\vec{F} = m\vec{a}$$
$$\vec{F_P} + \vec{F_E} = m \cdot \vec{a}$$
$$-mg - E|e| = m \cdot a \rightsquigarrow a = -g - \frac{Q}{\varepsilon_0 A}\frac{|e|}{m}$$

Per un condensatore piano $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{A}\frac{1}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$

$$y(t) = V_0 t - \frac{1}{2}\left(g + \frac{Q|e|}{\varepsilon_0 A \cdot m_e}\right) t^2$$

$$v(t) = V_0 - \left(g + \frac{Q|e|}{\varepsilon_0 A \cdot m_e}\right) t$$

$$\rightsquigarrow \begin{cases} \frac{h}{2} = V_0 t - \frac{1}{2} a t^2 \\ 0 = V_0 - a t \end{cases}$$

Con la conservazione energia:

Inizio: $K_i + U_{i,elett} + U_{i,grav} = \frac{1}{2} m_e V_0^2 + 0 + 0$

Fine: $K_f + U_{f,elett} + U_{f,grav} = 0 + E|e|\frac{h}{2} + m_e g \frac{h}{2}$